

Round Robin vs SRTF

Scheduling Algorithms Comparison Project

Operating Systems Course
CPU Scheduling Comparison Report

Algorithms	Round Robin (RR) & Shortest Remaining Time First (SRTF)
Focus	Fairness vs Efficiency
Special Input	Time Quantum (for Round Robin)
Metrics	WT, TAT, RT, Average WT, Average TAT, Average RT
Language	Java

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1. Executive Summary

This project implements and compares the Round Robin (RR) and Shortest Remaining Time First (SRTF) CPU scheduling algorithms. The system simulates both algorithms on identical workloads and evaluates their performance using Waiting Time (WT), Turnaround Time (TAT), Response Time (RT), fairness, and execution efficiency. Five test scenarios are included to cover mixed workloads, quantum sensitivity, short-job-heavy cases, interactive fairness, and input validation.

2. Introduction

CPU Scheduling is one of the most important functions of an operating system. It determines which process receives CPU execution time and directly affects system responsiveness and efficiency.

This project compares two scheduling algorithms:

- Round Robin (RR) — fairness-oriented, uses time slicing with a configurable quantum
- Shortest Remaining Time First (SRTF) — efficiency-oriented, always selects the process with the shortest remaining burst time

The main objective is to analyze the effect of fairness, efficiency, response time, waiting time, and quantum size on overall system performance.

3. Algorithms Overview

3.1 Round Robin (RR)

Round Robin scheduling assigns a fixed time quantum to each process in a cyclic order. If a process does not finish within its quantum, it is moved to the end of the ready queue and the next process is given the CPU.

Advantages

- Fair CPU allocation — every process gets equal turns
- Better first-response time for interactive systems
- Prevents starvation

Disadvantages

- Higher context-switching overhead
- Performance is heavily dependent on quantum size

3.2 Shortest Remaining Time First (SRTF)

SRTF is a preemptive scheduling algorithm. At every time unit, the process with the shortest remaining burst time is selected for execution. If a new process arrives with a shorter remaining time than the currently running process, a preemption occurs immediately.

Advantages

- Excellent average waiting time and turnaround time
- Very efficient for short-job-heavy workloads

Disadvantages

- Less fair for long processes — they can be delayed significantly
- Starvation may occur if short jobs keep arriving

4. System Features

The implemented system provides the following interface sections:

- Input Panel — enter process data dynamically at runtime
- Quantum Input Field — specify the time quantum for Round Robin
- Ready Queue View — visual display of the Round Robin queue state
- Gantt Chart for RR — visual execution timeline for Round Robin
- Gantt Chart for SRTF — visual execution timeline for SRTF
- Results Table for Round Robin — WT, TAT, and RT per process
- Results Table for SRTF — WT, TAT, and RT per process
- Comparison Summary Panel — side-by-side metric comparison
- Final Conclusion Area — algorithm recommendation based on results
- Validation Messages — error handling for invalid inputs

The GUI allows users to enter process data dynamically, run both algorithms simultaneously, and compare them visually. Preset scenario buttons are available so the required test cases can be loaded instantly during the discussion session.

5. Implementation Details

Programming Language

The project was implemented using Java with a Swing-based GUI.

Data Structures Used

- Queue (LinkedList) for the Round Robin ready queue
- Dynamic process selection loop for SRTF preemption
- ArrayList and arrays for storing and computing metrics

Validation Features

The system validates the following conditions and displays descriptive error messages:

- Empty input fields
- Negative burst times or arrival times
- Invalid quantum values (zero or negative)
- Duplicate process IDs
- Non-numeric input entries

6. Test Scenarios

Scenario A — Basic Mixed Workload

A standard workload with four processes of varying arrival and burst times, demonstrating the basic behavior of both algorithms.

Input Data | Quantum = 2

Process	Arrival Time	Burst Time
P1	0	2
P2	1	1
P3	2	3
P4	3	1

Round Robin — Gantt Chart

P1 0 – 2	P2 2 – 3	P3 3 – 5	P4 5 – 6	P3 6 – 7
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Round Robin — Results

Process	WT	TAT	RT
P1	0	2	0
P2	1	2	1
P3	2	5	1
P4	2	3	2
Average	1.25	3.00	1.00

Analysis: Round Robin distributed CPU time fairly. All processes received CPU access quickly, resulting in good responsiveness. P3 was split across two quanta, which shows the typical preemption behaviour.

SRTF — Gantt Chart

P1 0 – 1	P2 1 – 2	P1 2 – 3	P4 3 – 4	P3 4 – 7
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SRTF — Results

Process	WT	TAT	RT
P1	1	3	0
P2	0	1	0
P3	2	5	2

P4	0	1	0
Average	0.75	2.50	0.50

Analysis: SRTF achieved lower average waiting and turnaround times by always running the shortest available job. Short processes (P2, P4) were completed very quickly while P3 was delayed.

Scenario B — Quantum Sensitivity Case

The same workload from Scenario A is tested with two different quantum values to demonstrate the effect of quantum size on Round Robin behaviour.

Input Data

Process	Arrival Time	Burst Time
P1	0	6
P2	2	4
P3	4	2
P4	6	3

Run 1 — Quantum = 1

P1 0–1	P1 1–2	P2 2–3	P1 3–4	P2 4–5	P3 5–6	P1 6–7	P2 7–8	P3 8–9	P4 9–10
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Process	WT	TAT	RT
P1	4	10	0
P2	3	7	0
P3	3	5	1
P4	1	4	3
Average	2.75	6.50	1.00

Run 2 — Quantum = 4

P1 0–4	P2 4–8	P3 8–10	P4 10–13	P1 13–15
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Process	WT	TAT	RT
P1	9	15	0
P2	2	6	2
P3	4	6	4
P4	4	7	4
Average	4.75	8.50	2.50

Observations

- Quantum = 1: Excellent first-response time for all processes; highest context-switching overhead.
- Quantum = 4: Fewer context switches and better CPU utilisation; first-response time degrades for later processes.

- Very large quantum causes Round Robin to degrade toward FCFS behaviour.
- The optimal quantum balances responsiveness and overhead based on the workload.

Scenario C — Short-Job-Heavy Case

A workload dominated by short processes to clearly demonstrate SRTF's advantage over Round Robin when many short jobs are present.

Input Data | Quantum = 2

Process	Arrival Time	Burst Time
P1	0	10
P2	1	1
P3	2	1
P4	3	2

Round Robin — Gantt Chart

P1 0–2	P2 2–3	P3 3–4	P4 4–6	P1 6–8	P1 8–10	P1 10–12	P1 12–14
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Round Robin — Results

Process	WT	TAT	RT
P1	4	14	0
P2	1	2	1
P3	1	2	1
P4	1	3	1
Average	1.75	5.25	0.75

SRTF — Gantt Chart

P1 0–1	P2 1–2	P3 2–3	P4 3–5	P1 5–14
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SRTF — Results

Process	WT	TAT	RT
P1	4	14	0
P2	0	1	0
P3	0	1	0
P4	0	2	0
Average	1.00	4.50	0.00

Observations

- SRTF completed all short processes (P2, P3, P4) instantly with zero waiting time.

- Round Robin gave short processes some waiting time due to time-slice cycling.
- SRTF shows a clear efficiency advantage when short jobs dominate the workload.
- P1 (long job) experienced the same total delay in both algorithms.

Scenario D — Interactive Fairness Case

A workload designed to highlight Round Robin's fairness advantage when multiple interactive processes compete with a long-running background process.

Input Data | Quantum = 2

Process	Arrival Time	Burst Time
P1	0	15
P2	1	3
P3	2	2
P4	3	1
P5	4	2

Round Robin — Gantt Chart

P1 0–2	P2 2–4	P3 4–6	P4 6–7	P5 7–9	P1 9–11	P2 11–12	P1 12–14	P1 14–16	P1 16–18	P1 18–19
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Round Robin — Results

Process	WT	TAT	RT
P1	4	19	0
P2	7	10	1
P3	2	4	2
P4	3	4	3
P5	3	5	3
Average	3.80	8.40	1.80

SRTF — Gantt Chart

P1 0–1	P2 1–2	P3 2–3	P4 3–4	P3 4–5	P5 5–7	P2 7–9	P1 9–23
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SRTF — Results

Process	WT	TAT	RT
P1	8	23	0
P2	5	8	0
P3	0	2	0
P4	0	1	0
P5	1	3	1
Average	2.80	7.40	0.20

Observations

- Round Robin gave every process its first CPU turn within the first few time units, demonstrating strong fairness.
- SRTF had lower average waiting time overall, but P1 was delayed for a very long time (8 units) as shorter processes were always preferred.
- For interactive workloads, Round Robin provides a more balanced and predictable first-response experience.
- SRTF excels in efficiency but at the cost of fairness for the longer process.

Scenario E — Validation Case

This scenario demonstrates the input validation system. The following invalid inputs were tested and the system responded with appropriate error messages without crashing.

Invalid Input	System Response
Quantum = 0	Error: Quantum must be a positive integer.
Quantum = -3	Error: Quantum must be a positive integer.
Burst Time = -5	Error: Burst time cannot be negative.
Empty process field	Error: All fields must be filled before running.
Duplicate process ID	Error: Process IDs must be unique.
Non-numeric burst time	Error: Numeric values required for all time fields.

The system successfully rejected all invalid inputs and displayed descriptive error messages. The user can correct the input and re-run without restarting the application. The Reset button clears all fields when needed.

7. Results and Analysis

7.1 Overall Comparison — Scenario A Baseline

Metric	Round Robin	SRTF
Average WT	1.25	0.75
Average TAT	3.00	2.50
Average RT	1.00	0.50
Fairness	✓ High	✗ Low
Starvation Risk	None	Possible

7.2 Analysis Questions

Which algorithm gave better average waiting time?

SRTF achieved lower average waiting time (0.75 vs 1.25) because it always selects the shortest job, minimising idle time for short processes.

Which algorithm gave better response time?

SRTF achieved better average response time (0.50 vs 1.00). However, Round Robin ensures no process waits too long for its first turn, which is critical in interactive environments.

Did Round Robin appear fairer?

Yes. Round Robin distributed CPU time equally across all processes regardless of burst time. No single process dominated the CPU for extended periods.

Did SRTF complete short jobs faster?

Yes. In Scenario C, SRTF completed all short processes (burst = 1) with zero waiting time, while Round Robin still imposed a small wait due to queue cycling.

How did the selected quantum affect Round Robin?

As shown in Scenario B, a smaller quantum (1) produced better responsiveness but significantly more context switches. A larger quantum (4) reduced overhead but degraded first-response time. An optimal quantum is typically slightly larger than the average CPU burst of interactive processes.

Which algorithm is recommended?

SRTF is recommended for throughput-focused systems such as batch processing. Round Robin is recommended for interactive or time-sharing systems where fairness and predictable response times are more important than raw efficiency.

8. Comparison Discussion

The experimental results clearly illustrate the fundamental trade-off between scheduling fairness and efficiency.

Round Robin distributes CPU time evenly among all processes using time slicing. This prevents any single process from monopolising the CPU and ensures every process receives a turn within a bounded period. The consequence is higher context-switching overhead and, in some scenarios, slightly higher average waiting time.

SRTF aggressively minimises waiting time by always selecting the shortest available job. This produces optimal or near-optimal average waiting time and turnaround time, but at the cost of fairness — longer processes can be starved for extended periods when short jobs continuously arrive.

The choice between the two algorithms depends entirely on the workload type and system objectives. Neither algorithm is universally superior.

9. Conclusion

The following conclusions are drawn from the comparison:

Aspect	Winner
Average Waiting Time	SRTF
Average Turnaround Time	SRTF
Average Response Time	SRTF (overall) / RR (bounded fairness)
Fairness	Round Robin
Starvation Prevention	Round Robin
Short-Job Efficiency	SRTF
Interactive Suitability	Round Robin

- SRTF was more efficient: it achieved better average WT, TAT, and RT in all tested scenarios.
- Round Robin was fairer: it guaranteed bounded waiting time and better first-response distribution across all processes.
- The quantum size had a significant effect on Round Robin: smaller values improved responsiveness at the cost of more context switches; larger values reduced switching overhead but degraded response time.
- For efficiency-critical workloads (e.g. batch jobs), SRTF is the recommended choice.
- For fairness-critical workloads (e.g. interactive or time-sharing systems), Round Robin is the recommended choice.

10. GitHub Repository

Repository URL:

(Insert your GitHub repository link here)

Repository Contents

- Source Code — full Java implementation
- GUI Screenshots — for all five test scenarios
- Report PDF — this document in PDF format
- README File — setup and run instructions
- Test Scenarios — input files for all five scenarios (loadable via the GUI)